

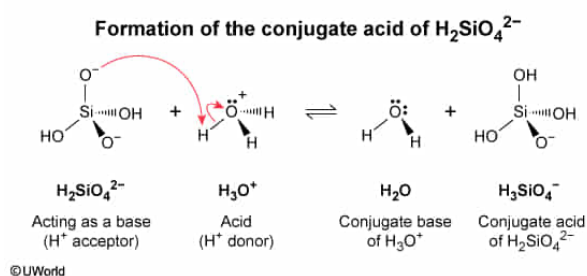
The conjugate acid of the dihydrogen orthosilicate anion ($\text{H}_2\text{SiO}_4^{2-}$) is:

- ☐ A. HSiO_4^{3-} . [12%]
- ✓ ☒ B. H_3SiO_4^- . [78%]
- ☐ C. H_4SiO_4 . [5%]
- ☐ D. H_3O^+ . [3%]

Correct

79% answered correctly

Explanation:



According to the **Brønsted-Lowry** definitions, an acid is a chemical species capable of donating a proton (H^+ ion), and a base is a chemical species that can accept a proton. Acids that can donate more than one proton are called **polyprotic acids**. Such acids donate protons in a **stepwise** manner and produce more than one ionic species.

The **conjugate** of an acid (or base) is the species that the acid (or base) becomes after losing (or gaining) one proton. A base is converted into its **conjugate acid** when the base accepts a proton. Likewise, an acid is converted into its **conjugate base** when the acid donates a proton. Therefore, an acid (or base) and its conjugate have chemical formulas that differ by only one proton.

With two acidic protons and two negative charges, the $\text{H}_2\text{SiO}_4^{2-}$ anion can function as either an acid (proton donor) or a base (proton acceptor). If $\text{H}_2\text{SiO}_4^{2-}$ acts as a base and accepts a proton, an ion of H_3SiO_4^- is formed. Because the newly formed H_3SiO_4^- anion has one more available proton to donate compared with $\text{H}_2\text{SiO}_4^{2-}$, H_3SiO_4^- is the conjugate acid of $\text{H}_2\text{SiO}_4^{2-}$.

(Choice A) HSiO_4^{3-} is the conjugate *base*, rather than the conjugate acid, of $\text{H}_2\text{SiO}_4^{2-}$. Donation of a proton by $\text{H}_2\text{SiO}_4^{2-}$ (acting as an acid) forms an ion of

HSiO_4^{3-} , which has one less acidic proton than $\text{H}_2\text{SiO}_4^{2-}$.

(Choice C) Although the stepwise loss of two protons does produce $\text{H}_2\text{SiO}_4^{2-}$, the chemical formula H_4SiO_4 differs by more than one proton from $\text{H}_2\text{SiO}_4^{2-}$. Therefore, H_4SiO_4 is not the conjugate acid of $\text{H}_2\text{SiO}_4^{2-}$.

(Choice D) The hydronium ion (H_3O^+) forms when an acid donates a proton to a water molecule (H_2O acts as a base). H_3O^+ differs by more than a proton from $\text{H}_2\text{SiO}_4^{2-}$ and is not the conjugate acid of $\text{H}_2\text{SiO}_4^{2-}$.

Educational objective:

The conjugate of an acid (or base) is the species that the acid (or base) becomes after losing (or gaining) one proton. An acid (or base) and its conjugate have chemical formulas that differ by only one proton.

Time spent:0 sec

Subject:

General Chemistry

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System:

Solutions and Electrochemistry